



Loitering munition

A **loitering munition**, also known as a **suicide drone**,^{[1][2][3][4]} **kamikaze drone**,^{[5][6][7]} or **exploding drone**,^[8] is a weapon with a warhead that is typically designed to loiter until a target is designated, then crash into it.^{[9][10][11]} They enable attacks against hidden targets that emerge for short periods without placing high-value platforms near the target area. Unlike many other types of munitions, their attacks can be changed mid-mission or aborted. Loitering munitions are typically aerial platforms, but include some autonomous undersea vehicles with similar characteristics.^[12]

Loitering weapons emerged in the 1980s for the Suppression of Enemy Air Defenses (SEAD) role, and were deployed for SEAD by some military forces in the 1990s. In the 2000s, they were developed for additional roles, from long-range strikes and fire support to short-range tactical systems that fit in a backpack.

History

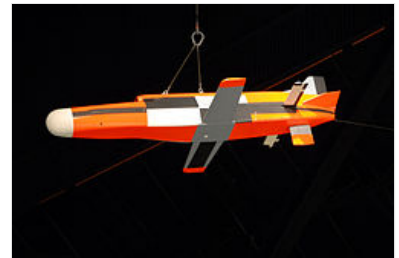
First development and terminology

Initially, loitering munitions were not referred to as such but rather as "suicide UAVs" or "loitering missiles". Different sources point at different projects as originating the weapon category. The failed US AGM-136 Tacit Rainbow program^{[13][14]} or the 1980s initial Israeli Delilah variants^{[15][16]} are mentioned by some sources.^[17] The Iranian Ababil-1 was produced in the 1980s but its exact production date is unknown.^[18] The Israeli IAI Harpy was produced in the late 1980s.^[17]

Early projects did not use the "loitering munition" nomenclature, which emerged much later; they used terminology existing at the time. For instance the AGM-136 Tacit Rainbow was described in a 1988 article:

the Tacit Rainbow unmanned jet aircraft being developed by Northrop to loiter on high and then swoop down on enemy radars could be called a UAV, a cruise missile, or even a standoff weapon. But it is most definitely not an RPV.

—Canan, James W. "Unmanned Aerial Vehicles." (<http://www.airforcemag.com/MagazineArchive/Documents/1988/October%201988/1088uav.pdf>) *Air Force Magazine* (1988), page 87



Northrop AGM-136 Tacit Rainbow on display at the National Museum of the U.S. Air Force in Dayton, Ohio.



IAI Harpy first-generation loitering munition for SEAD role

Initial role in suppression of enemy air defense

In response to the first generation of fixed-installation surface-to-air missiles (SAMs) such as S-75 and S-125, the U.S. military developed SEAD doctrine and Wild Weasel weapons, including anti-radiation missiles (ARMs) such as AGM-45 Shrike. The Soviet Union countered with mobile SAMs such as 2K12 Kub and intermittent use of radar.^[19] In Israel's 1982 Operation Mole Cricket 19, UAVs and air-launched Samson decoys were used over suspected SAM areas to saturate enemy SAMs and to bait them to activate their radar systems, which were then attacked by ARMs.^{[20][21]}

In the 1980s, programs such as the IAI Harpy and AGM-136 Tacit Rainbow integrated anti-radiation sensors into a drone or missile coupled with command and control and loitering capabilities. This allowed the attacking force to put relatively cheap munitions over suspected SAM sites, then attack when the SAM battery was spotted.^{[22][23][24][25]}



Loitering Munitions HERO (UVision Air Ltd, Israel), DSEI 2019, London.

Evolution into additional roles

Starting in the 2000s, loitering weapons have been developed for additional roles from relatively long-range strikes and fire support^[26] to tactical, very-short-range tactical use.^{[27][28][29][30]} In the 2016 Nagorno-Karabakh conflict, an IAI Harop was used against a bus used as a troop transport for Armenian soldiers.^[7] The ZALA Lancet and several Shahed drones, including the HESA Shahed 136, have been used by

Russia in the Russian invasion of Ukraine, while Ukraine has fielded loitering munitions such as the UJ-25 Skyline and the American-made AeroVironment Switchblade, which is deployed to platoons and fits in a backpack.

During conflicts in the 2010s and 2020s, conventional armies and non-state militants began modifying common commercial racing drones into "FPV loitering munitions" by the attachment of a small explosive, so-named because of the first-person view (FPV) they provide the operator. Explosive ordnance such as an IED, grenade, mortar round or an RPG warhead are fitted to an FPV drone then deployed to aerial bomb tactical targets. FPV drones also allow direct reconnaissance during the drone's strike mission.^{[31][32]}

After the Russian invasion of Ukraine began in 2022, both Russian and Ukrainian forces were producing thousands of FPV drones every month by October 2023, many of which were donated by volunteer groups.^[33] Escadrone Pegasus and the Vyriy Drone Molfar are two examples of the low-cost drones that rapidly evolved in 2022–23 during the war.^[34] In 2022, the UK Government announced it was providing "hundreds of loitering munitions" to Ukraine.^[35] On 9 November 2023, Ukrainian soldiers claimed to have used a civilian-donated FPV drone to destroy a Russian Tor missile system on the Kupiansk front, showcasing the potential cost-effectiveness of fielding such munitions. A Tor missile system costs some \$24 million dollars to build, which could buy 14,000 FPV drones.^{[36][37]}



XM501 US prototype capable of launching LAM (loitering attack munition).

Characteristics

Loitering munitions may be as simple as an unmanned aerial vehicle (UAV) with attached explosives that is sent on a potential *kamikaze* mission, and may even be constructed with commercially available quadcopters with strapped-on explosives.^[38]

Purpose-built munitions are more elaborate in flight and control capabilities, warhead size and design, and onboard sensors for locating targets.^[39] Some loitering munitions use a human operator to locate targets whereas others, such as IAI Harop, can function autonomously searching and launching attacks without human intervention.^{[40][41]} Another example is UVision HERO solutions – the loitering systems are operated remotely, controlled in real time by a communications system and equipped with an electro-optical camera whose images are received by the command and control station.^{[42][43]}



Air-launched Delilah loitering munition, controlled by backseat WSO

Some loitering munitions may return and be recovered by the operator if they are unused in an attack and have enough fuel; in particular this is characteristic of UAVs with a secondary explosive capability.^[44] Other systems, such as the Delilah^{[15][45][11]} do not have a recovery option and are self-destructed in mission aborts.

Countermeasures

Russia uses ZALA Lancet drones in Ukraine. Since spring 2022 Ukrainian forces have been building cages around their artillery pieces using chain link fencing, wire mesh and even wooden logs as part of the construction. One analyst told Radio Liberty that such cages were "mainly intended to disrupt Russian Lancet munitions." A picture supposedly taken from January 2023 shows the rear half of a Lancet drone that failed to detonate due to such cages. Likewise Ukrainian forces have used inflatable decoys and wooden vehicles, such as HIMARS, to confuse and deceive Lancet drones.^{[46][47]}

Ukrainian soldiers report shooting down Russian drones with sniper rifles.^[48] Russian soldiers use electronic warfare to disable or misdirect Ukrainian drones and have reportedly used the Stupor anti-drone rifle, which uses an electromagnetic pulse that disrupts a drone's GPS navigation.^[49] A Royal United Services Institute study in 2022 found that Russian Electronic Warfare units, in March and April 2022, knocked out or shot down 90% of Ukrainian drones that they had at the start of the war in February 2022. The main success was in jamming GPS and radio links to the drones.^[50]

Both Ukraine and Russia rely on electronic warfare to defeat FPV drones. Such jammers are now used on Ukrainian trenches and vehicles.^[51] Russian forces have built jammers that can fit into a backpack.^[52] Pocket-size jammers for soldiers were also developed.^[53] As of June 2023 Ukraine was losing 5-10,000 drones a month, or 160 per day, according to Ukrainian soldiers.^[54]

This has led to Russia creating wire guided FPV drones, similar to a wire-guided missile or even wire-guided torpedoes. Those drones typically have fibre optic cables 5-20 km in length. Such guidance makes the link between operators and FPV drone immune to jamming.^[55] It also allows for much faster and better quality updates from the drone, even from locations where radio contact would be poor, and doesn't reveal operator's or drone's location by radio signals.^{[56][57]} They also need less power to communicate, and so can be used to idle on the ground for ambushes.^[58] They have reduced range, payload and manoeuvrability compared to wireless drones,^{[59][60]}

although in practice, range and agility of the wired drones can be even higher than those of the radio-controlled ones, given their reduced control latency and increased survivability.^[61] Ukraine has also responded by using autonomous drones tasking to ensure that a jammed drone can hit a target. In March 2024 footage put on social media showed a Ukrainian FPV drone being jammed just before it struck a target. Despite the loss of operator control it still managed to strike the target.^[62]

Russian tanks have been fitted with rooftop slat armor at the beginning of the Russian invasion of Ukraine which could provide protection against loitering munitions in some circumstances. Some Ukrainian tanks taking part in the 2023 Ukrainian counteroffensive were also spotted using roof screens.^{[63][64][65]}

On 21 March 2024, recent footage of the submarine Tula showed that it has been fitted with a slat armor to prevent drone strikes, the first ocean-going asset to carry such a modification.^[66]

Comparison to similar weapons

Loitering munitions fit in the niche between cruise missiles and unmanned combat aerial vehicles (UCAVs).^{[11][67]}

The following table compares similar size-class cruise missiles, loitering munitions, and UCAVS:

Characteristic	Cruise missile	Loitering munition	UCAV
Cost appropriate for expendable one-time use	Yes	Yes	No, but high cost allows for higher-quality platform
Recovery possible after launch	No	Sometimes	Yes, typical mission profile is round-trip
Built-in warhead	Yes	Yes	No
Stealthy final dive to target	Usually yes	Usually yes	Usually no
Loitering	No or limited	Yes	Usually yes
Sensors for target acquisition	Limited	Yes	Usually yes
Command and control during flight	Usually limited	Yes	Yes
Range	~1500km ^[68]	Varies, from 10-1000km	~1000km
Speed	Typically higher	Typically lower	Depends on role
Examples			
Example type	 Block IV Tomahawk cruise missile. Its small wing area is optimized for high-speed cruise.	 Static display of an IAI Harop loitering munition optimized for the <u>Suppression of Enemy Air Defenses (SEAD)</u> role	 <u>General Atomics MQ-1 Predator UCAV</u>
Range	1,600 km	1,000 km	1,100 km
Max speed	high sub-sonic, 880 km/h	190 km/h	217 km/h
Flight endurance	c. 2 hours	6 hours	24 hours
Engine	3.1 kilonewtons (700 lbf) F107-WR-402 turbofan engine	37 hp (28 kW) <u>Wankel</u> engine	115 hp (86 kW) <u>Rotax 914F</u>
Loaded system weight	1,588 kg	135 kg	1,020 kg
Payload	450 kg warhead	23 kg warhead	up to 204 kg (2 × AGM-114 Hellfire or 6 × AGM-176 <u>Griffin</u> air-to-surface missiles)
Length	6.25 m	2.5 m	8.22 m
Wingspan	2.67 m ^[69]	3 m ^[39]	16.8 m ^[70]

Whereas some cruise missiles, such as the Block IV Tomahawk, have the ability to loiter and have some sensory and remote control features,^[71] their primary mission is typically strike and not target acquisition. Cruise missiles, as their name implies, are optimized for long-range flight at constant speed both in terms of propulsion systems and wings or lifting body design. They are often unable to loiter at slow fuel-efficient speeds which significantly reduces potential loiter time even when the missile has some loiter capabilities.^[72]

Conversely almost any UAV could be piloted to crash onto a target and most could be fitted with an improvised explosive warhead.^[38] However the primary use of a UAV or UCAV would be for recoverable flight operations carrying reconnaissance equipment and/or munitions. While many UAVs are explicitly designed with loitering in mind, they are not optimized for a diving attack, often lacking forward facing cameras, lacking in control response-speed which is unneeded in regular UAV flight, and are noisy when diving, potentially providing warning to the target. UAV's, being designed as multi-use platforms, often have a unit cost that is not appropriate for regular one-time expendable mission use.^{[73][67]}

The primary mission of a loitering munition is reaching the suspected target area, target acquisition during a loitering phase, followed by a self-destructive strike, and the munition is optimized in this regard in terms of characteristics (e.g. very short engine lifetime, silence in strike phase, speed of strike dive, optimization toward loitering time instead of range/speed) and unit cost (appropriate for a one-off strike mission).^{[74][75]}



NCSIST Chien Hsiang, an example of an expendable loitering munition

Ethical and international humanitarian law concerns

Loitering munitions that can make autonomous attack decisions (man out of the loop) raise moral, ethical, and international humanitarian law concerns because a human being is not involved in the decision to attack and potentially kill humans. A distinction is often drawn with fire-and-forget missiles in common use since the 1960s, which may lock-on after launch or be sensor-fuzed, but whose flight time is typically limited and a human launches them at an area where enemy activity is strongly suspected. An autonomous loitering munition, on the other hand, may be launched at an area where enemy activity is only probable, and loiter searching autonomously for targets for hours after the initial launch decision, though it may be able to request final authorization for an attack from a human. The IAI Harpy and IAI Harop are frequently cited as aerial systems that set a precedent in this way—though some note that naval mines also loiter and may kill indiscriminately.^{[76][77][78][79][80][81]}

Users and producers

As of 2023, loitering munitions are used by the armed forces of several countries, including:

-  Albania – YIHA-III
-  Argentina – HERO 30, HERO 120^[82]
-  Armenia – HRESH, BEEB 1800,^[83] AW21^[84]
-  Australia – Drone 40,^[85] Innovaero OWL
-  Azerbaijan – IAI Harpy, IAI Harop, Orbiter 1K,^[7] SkyStriker, STM Kargu,^{[86][87]} Qirği, Quzgün^[88]
-  Belarus – UBAK-25 Chekan^[89]
-  Brazil – Anshar^[90]
-  China – IAI Harpy, CH-901, WS-43, ASN-301^{[91][92]}
-  France - Switchblade, Colibri, Larinae
-  Georgia - Delta-WB Warmate^{[93][94]}
-  Greece – Attalus,^[95] Aihmi AHM-1X^[96]
-  Indonesia – Rajata,^[97] Minibe^[98]
-  India – Nagastra-1,^[99] IAI Harpy, IAI Harop,^{[100][101]} SkyStriker (<https://elbitsystems.com/product/skystriker/>), Warmate, Trinetra, ALS-50,^[102] Johnnette JM-1 (<https://www.johnnette.com/jm1.html>), Shaurya-1, Kadet (<http://www.kadet-uav.com/>) Loitering Aerial Munition,^[103] Overwatch PHOLOS, Zulu DRAP, Sureshastra Mk1
-  Iran – Karrar, Shahed 131, Shahed 136 (loitering capabilities disputed),^{[104][105]} Hesa Ababil-2, Raad 85, Arash-2, Meraj-521, Meraj-532, Zhubin, Shahin-1, Shahed 238 and possibly others^{[106][107][108][109]}
-  Israel – IAI Harpy, IAI Harop, IAI Harpy NG, IAI Green Dragon, IAI Rotem L, Orbiter 1K,^[110] Delilah, SkyStriker,^[111] Spike Firefly, HERO loitering munitions series,^{[112][113][114][115]} Viper,^[116] Lanus,^[117] Point Blank,^[118] SpyX,^[119] and upgraded variants.^[120]
-  Lithuania – Switchblade^[121]
-  Morocco – IAI Harop, IAI Harpy, SpyX
-  Pakistan – E- RAD, Rover LM, Xpear MX-150, NASTP Dark Angel Series, YIHA-III, GIDS Blaze series, NASTP KaGeM V3
-  Poland – WB Electronics Warmate^[122]
-  Portugal – UAVision Elanus^[123]
-  Russia – ZALA Kub-BLA ("Cube"), ZALA Lancet,^{[124][125][126]} Geran-1, Geran-2, Ushkuynik KVN
-  Serbia – Gavran,^[127] Osica,^[128] Komarac,^[129] Vila 1^[130]
-  Singapore – IAI Harop
-  Slovakia – AX-2 Predator
-  South Africa – Paramount N-Raven^[131]
-  South Korea – Devil Killer,^{[132][133]} IAI Harpy
-  Spain - Q-SLAM-40
-  Sudan – Kamin-25
-  Taiwan – NCSIST Chien Hsiang,^[134] NCSIST Fire Cardinal

-  Turkey - Robit UAV AZAB,^[135] IAI Harpy,^[136] STM Kargu,^[137] STM Alpagu,^[138] Transvaro-Havelsan Fedai,^[139] LENTATEK Kargi,^[140] Roketsan-STM Alpagut^{[141][142]}
-  Turkmenistan – SkyStriker
-  UAE – QX-1,^{[143][144]} Hunter SP,^[145] Hunter 2-S,^{[146][147]} Hunter 5, Hunter 10,^[148] Shadow 25, Shadow 50,^{[149][150]} RW-24,^[151] N-Raven^[152]
-  United Kingdom – Switchblade, Overwatch PHOLOS
-  United States – ALTIUS-600M, AeroVironment Switchblade, Phoenix Ghost, Raytheon Coyote,^[153] HERO 120, Point Blank, Northrop Grumman Lumberjack.^[154]
-  Ukraine – ALTIUS-600M, AQ-400 Scythe, Bober, Liutyi,^[155] Overwatch PHOLOS, Phoenix Ghost, QinetiQ Banshee, RAM II, ST-35 Silent Thunder, Switchblade, UJ-25 Skyline, Warmate
-  Yemen – (Houthis) – Qasef-1/2K,^[156] Shahed 131, Shahed 136, Samad-2/3, Shahed-101

North Korea

In March 2025, North Korean state media said leader Kim Jong Un oversaw testing of AI-equipped reconnaissance and suicide drones produced by the country's Unmanned Aerial Technology Complex, inspected a new reconnaissance drone and an early warning and control (AEW) aircraft, and pushed to expand the production of unmanned systems, citing their importance in modern warfare.^[157]

See also

- Boeing Persistent Munition Technology Demonstrator
- Copperhead (UUV), a family of autonomous unmanned undersea vehicles with loitering munitions capability.^[12]
- Flying bomb
- Low Cost Autonomous Attack System
- Sypaq Corvo Precision Payload Delivery System
- Television guidance
- V-1 flying bomb
- XM501 Non-Line-of-Sight Launch System

References

1. US army may soon use Israeli-designed 'suicide drones' (<http://www.jpost.com/Israel-News/US-army-may-use-soon-Israeli-designed-suicide-drones-455572>), *Jerusalem Post*, June 2016
2. China Unveils a Harpy-Type Loitering Munition (<http://www.israeldefense.co.il/en/node/28716>), Israel Defense, March 2017
3. Rogoway, Tyler (8 August 2016). "Meet Israel's "Suicide Squad" of Self Sacrificing Drones" (<https://www.twz.com/4760/mee-t-israels-suicide-squad-of-self-sacrificing-drones>). *The War Zone*. Retrieved 18 December 2024.
4. Loitering Munitions – In Focus (<http://dronecenter.bard.edu/files/2017/02/CSD-Loitering-Munitions.pdf>), Center for the Study of the Drone, Feb 2017
5. Kamikaze drone loiters above, waits for target (<https://www.cnet.com/news/kamikaze-drone-loiters-above-waits-for-target/>), CNET, June 2009
6. 'Kamikaze drones' add a new layer of lethality to remote force (<http://www.c4isrnet.com/story/military-tech/omr/missile-defense/2015/08/13/kamikaze-drones-add-new-layer-lethality-remote-forces/31405521/>) Archived (<https://web.archive.org/web/20150919193839/http://www.c4isrnet.com/story/military-tech/omr/missile-defense/2015/08/13/kamikaze-drones-add-new-layer-lethality-remote-forces/31405521/>) 19 September 2015 at the Wayback Machine, C4ISRNET, August 2015
7. Israeli-made kamikaze drone spotted in Nagorno-Karabakh conflict (<https://www.washingtonpost.com/news/checkpoint/wp/2016/04/05/israeli-made-kamikaze-drone-spotted-in-nagorno-karabakh-conflict/>), *Washington Post*, April 2016
8. "Kyiv pummelled by Putin's exploding drones, Vitali Klitschko says" (<https://www.independent.co.uk/news/putin-kyiv-drone-strike-klitschko-b2254555.html>). *The Independent*. 2 January 2023.
9. Loitering Munition Availability Expanding Internationally (<http://aviationweek.com/defense/loitering-munition-availability-expanding-internationally>), *Aviation Week*, April 2016
10. Loitering Weapon Systems – A Growing Demand (<http://i-hls.com/archives/73521>), h-ils, December 2016
11. Watch This Drone Turn Into A Missile (<http://www.popsci.com/watch-drone-turn-missile>), *Popular Science*, August 2015
12. Trevithick, Joseph (7 April 2025). "Copperhead Torpedo-Like Underwater Kamikaze Drones Rolled Out By Anduril" (<https://www.twz.com/sea/copperhead-torpedo-like-underwater-kamikaze-drones-rolled-out-by-anduril>). *The War Zone*. Retrieved 10 April 2025.
13. An Introduction to Autonomy in Weapon Systems (https://www.files.ethz.ch/isn/188865/Ethical%20Autonomy%20Working%20Paper_021015_v02.pdf), pages 13–14, By: Paul Scharre and Michael C. Horowitz, CNAS Working paper, Feb 2015

14. Loitering munitions fit in the niche between cruise missiles and unmanned combat aerial vehicles (UCAVs or combat drones), sharing characteristics with both. They differ from cruise missiles in that they are designed to loiter for a relatively long time around the target area, and from UCAVs in that a loitering munition is intended to be expended in an attack and has a built-in warhead. As such, they can also be considered a nontraditional ranged weapon. Canan, James W. "Unmanned Aerial Vehicles." *Air Force Magazine* (1988). (<http://www.airforcemag.com/MagazineArchive/Documents/1988/October%201988/1088uav.pdf>), page 87
15. *The Secrets of Delialah* (Hebrew) (<http://www.iaf.org.il/2158-28495-he/IAF.aspx>), IAF bulletin, issue 184, December 2008
16. Loitering Weapons are making a Comeback (http://defense-update.com/20090614_loitering_weapons.html), Defense Update, June 2009
17. *Drone Strike!: UCAVs and Aerial Warfare in the 21st Century* (<https://books.google.com/books?id=DB7xDQAAQBAJ&dq=H+UAV&pg=PA106>), By Bill Yenne, ISBN 9781580072526, pages 106–107
18. "چهارمین قدرت بهیادی دنیا ؛ از سینما تا جهان نما" (<https://www.pahpad.com/fa/news/348/%da%86%d9%87%d8%a7%d8%b1%d9%85%db%8c%d9%86-%d9%82%d8%af%d8%b1%d8%aa-%d9%be%d9%87%d9%be%d8%a7%d8%af%db%8c-%d8%af%d9%86%db%8c%d8%a7-%d8%a7%d8%b2-%d8%b3%db%8c%d9%86%d9%85%d8%a7-%d8%aa%d8%a7-%d8%ac%d9%87%d8%a7%d9%86-%d9%86%d9%85%d8%a7>). 13. *موسسه پرنده های هدایت پذیر از دور*. August 2021.
19. Reedy, Edward K. (1 January 1987). "Radar ECCM Considerations and Techniques". In Eaves, Jerry L.; Reedy, Edward K. (eds.). *Principles of Modern Radar* (<https://archive.org/details/principlesmodern00eave>). Springer US. pp. 681 (<https://archive.org/details/principlesmodern00eave/page/n677>)–699. doi:10.1007/978-1-4613-1971-9_22 (https://doi.org/10.1007%2F978-1-4613-1971-9_22). ISBN 978-1-4612-9170-1.
20. *Six Days in June* (Hebrew) (<http://www.iaf.org.il/625-19927-he/IAF.aspx>), IAF bulletin, issue 145, June 2002
21. Surface to Air Missile Effectiveness in Past Conflicts (<http://www.ausairpower.net/APA-SAM-Effectiveness.html>), Technical Report APA-TR-2010-1001, Dr Carlo Kopp, AFAIAA, SMIEEE, PEng, October 2010
22. Meet Israel's 'Suicide Squad' of Self-Sacrificing Drones (<http://www.thedrive.com/the-war-zone/4760/meet-israels-suicide-squad-of-self-sacrificing-drones>), August 2016, The Drive
23. ADAPTIVE DISCRETE EVENT SIMULATION FOR ANALYSIS OF HARPY SWARM ATTACK (<http://apps.dtic.mil/dtic/tr/fulltext/u2/a552529.pdf>), Brandon J. Cobb, Naval Postgraduate School, Thesis, September 2011
24. "Suminsby, Robert E. Fear no Evil: Unmanned combat air vehicles for suppression of enemy air defenses. AIR UNIV MAXWELL AFB AL AEROSPACE STUDIES INST, 2002" (<https://apps.dtic.mil/sti/pdfs/ADA420687.pdf>) (PDF). Archived (<https://web.archive.org/web/20170412145128/http://www.dtic.mil/cgi-bin/GetTRDoc?AD=ADA420687>) from the original on 12 April 2017.
25. Loitering, Smart Cruise Missile Marketed to U.S. Navy (<http://www.nationaldefensemagazine.org/archive/2001/June/Pages/Loitering7024.aspx>) Archived (<https://web.archive.org/web/20170412062613/http://www.nationaldefensemagazine.org/archive/2001/June/Pages/Loitering7024.aspx>) 12 April 2017 at the Wayback Machine, National Defense, June 2001
26. British Army Sets to Deploy Fire Shadow Loitering Weapons to Afghanistan by Early 2012 (http://defense-update.com/20110914_fire-shadow-afghanistan.html) Defense Update, September 2011
27. Military Investigates Killer Drones That Can Fit in Rucksacks (<http://www.nationaldefensemagazine.org/archive/2011/July/Pages/MilitaryInvestigatesKillerDronesThatCanFitinRucksacks.aspx>) Archived (<https://web.archive.org/web/20111020170449/http://www.nationaldefensemagazine.org/archive/2011/July/Pages/MilitaryInvestigatesKillerDronesThatCanFitinRucksacks.aspx>) 20 October 2011 at the Wayback Machine National Defense, July 2011
28. A Tiny Missile That Waits Overhead—Silent, Patient and Deadly, WarIsBoring (<https://medium.com/war-is-boring/a-tiny-missile-that-waits-overhead-silent-patient-and-deadly-6f5a0093cab0>), January 2014
29. AeroVironment producing Switchblade missiles for Army (<http://www.upi.com/Defense-News/2016/10/03/AeroVironment-producing-Switchblade-missiles-for-Army/1391475510902/>) UPI, October 2016
30. Loitering Autonomous Weapons (http://defense-update.com/features/du-1-07/armedUAVs_8.htm) Archived (https://web.archive.org/web/20130205125341/http://defense-update.com/features/du-1-07/armedUAVs_8.htm) 2013-02-05 at the Wayback Machine, Defense Update, January 2007
31. "Why Ukraine's kamikaze racing drones are causing a buzz on and off the battlefield" (<https://www.abc.net.au/news/2023-04-01/fpv-racing-drone-kamikaze-attacks-ukraine-russia-war/102155702>). *Australian Broadcasting Corporation*. 31 March 2023. Retrieved 14 March 2023.
32. "Ukraine Racing Drone Converted Into Loitering Munition Makes Precision Strike Through Doorway" (<https://www.forbes.com/sites/davidhambling/2022/08/01/ukraine-racing-drone-converted-into-loitering-munition-makes-precision-strike-through-doorway/?sh=69a1be2f6157>). *Forbes*. 1 August 2022. Retrieved 14 March 2023.
33. "Russia Adds Thermal Imaging To FPV Kamikaze Drones" (<https://www.forbes.com/sites/davidhambling/2023/10/11/russia-adds-thermal-imaging-to-fpv-kamikaze-drones/?sh=25b26dae4f7b>). *Forbes*. 11 October 2023. Retrieved 19 November 2023.
34. Hambling, David (5 May 2023). "Drones: Ukraine's Escadrone On The Skill Of Flying FPV Kamikazes" (<https://www.forbes.com/sites/davidhambling/2023/05/05/pilots-not-dronesukraines-escadrone-on-the-skill-of-flying-fpv-kamikazes/?sh=32fb7e4c12b7>). *Forbes*. Retrieved 15 March 2023.
35. "UK to send scores of artillery guns and hundreds of drones to Ukraine" (<https://www.gov.uk/government/news/uk-to-send-scores-of-artillery-guns-and-hundreds-of-drones-to-ukraine>). *GOV.UK*. Retrieved 13 September 2024.
36. Malyasov, Dylan (9 November 2023). "Ukraine's tiny drone blows up Russian Tor air defense system" (<https://defence-blog.com/ukraines-tiny-drone-blows-up-russian-tor-air-defense-system/>). *Defense Blog*. Retrieved 10 November 2023.
37. Nikolov, Boyko (8 November 2023). "Ukrainian FPVs killed their most expensive Russian target – TOR SAM" (<https://bulgarianmilitary.com/2023/11/08/ukrainian-fpvs-killed-their-most-expensive-russian-target-tor-sam/>). *Bulgarian Military*. Retrieved 10 November 2023.
38. ISIS Using Kamikaze Drones in Iraq (<http://www.popularmechanics.com/military/weapons/a23350/isis-using-kamikaze-drones-in-iraq/>), *Popular Mechanics*, October 2016

39. iClean – Loitering Attack UCAV (http://www.dror-aero.com/ac/iclean/iclean_fr.pdf), Artzi Dror, Technion Institute of Technology, 2012
40. Israel Unveils Loitering Anti-Missile Drone (<http://defense-update.com/products/h/harop.html>) Archived (<https://web.archive.org/web/20180315003749/http://defense-update.com/products/h/harop.html>) 2018-03-15 at the Wayback Machine, Defense Update, 2009
41. Harpy Air Defense Suppression System (<http://defense-update.com/directory/harpy.htm>) Archived (<https://web.archive.org/web/20171211211849/http://defense-update.com/directory/harpy.htm>) 2017-12-11 at the Wayback Machine, Defense Update, 2006
42. "Loitering Munitions – High Precision Systems" (<https://web.archive.org/web/20220506190110/https://uvisionuav.com/our-technology/>). *UVision*. Archived from the original (<https://uvisionuav.com/our-technology/>) on 6 May 2022. Retrieved 1 September 2021.
43. "Unmanned Aerial Loitering Systems for Various Missions" (<https://uvisionuav.com/products/>). *UVision*. Retrieved 1 September 2021.
44. IAI's ROTEM – Tactical Multirotor Killer Drone (http://defense-update.com/20160216_rotem.html), Defense Update, 2016
45. Delilah – The IAF Loitering Missile (http://defense-update.com/20090614_delilah-loitering-missile.html), Defense Update, June 2009
46. Amos Chapple (18 April 2023). "The Makeshift Armor Of The Ukraine War" (<https://www.rferl.org/a/makeshift-armor-ukraine-war-invasion-mad-max/32369222.html>). *rferl.org*.
47. "Inflatable tanks and wooden HIMARS: Fake, but work exceptionally well" (<https://www.europeantimes.news/2023/04/inflatable-tanks-and-wooden-himars-fake-but-work-exceptionally-well/>). *europeantimes.news*. 19 April 2023. Retrieved 21 April 2023.
48. Mike Collett-White (24 March 2023). "Ukrainians say Russian drones pose growing threat, more Stingers needed" (<https://www.reuters.com/world/europe/ukrainians-say-russian-drones-pose-growing-threat-more-stingers-needed-2023-03-23/>). *reuters.com*. Retrieved 26 April 2023.
49. "How are 'kamikaze' drones being used by Russia and Ukraine?" (<https://www.bbc.com/news/world-62225830.amp>). *bbc.com*. 3 January 2023. Retrieved 26 April 2023.
50. David Axe (24 December 2022). "Russia's Electronic-Warfare Troops Knocked Out 90 Percent Of Ukraine's Drones" (<https://www.forbes.com/sites/davidaxe/2022/12/24/russia-electronic-warfare-troops-knocked-out-90-percent-of-ukraines-drones/?sh=685ea817575c>). *forbes.com*. Retrieved 26 April 2023.
51. "Ukrainian EW developer on anti-drone warfare – interview" (<https://english.nv.ua/nation/ukrainian-ew-developer-on-countering-russian-drones-interview-50389565.html>). *New Voice of Ukraine*. 3 February 2024. Retrieved 25 March 2024.
52. "Russia Deploys Radio-Jammers To Ground Ukraine's Drones. Just One Problem: The Jammers Don't Work" (<https://www.forbes.com/sites/davidaxe/2024/01/31/russia-deployed-radio-jammers-to-ground-ukraines-drones-just-one-problem-the-jammers-dont-work/?sh=51e3ae7320e6>). *Forbes*. 31 January 2024. Retrieved 25 March 2024.
53. Oleksandr Tartachnyi (21 March 2024). "The Invisible War: Inside the electronic warfare arms race that could shape course of war in Ukraine" (<https://kyivindependent.com/the-invisible-war-inside-the-electronic-warfare-arms-race-that-could-shape-course-of-the-war/>). *The Kyiv Independent*. Retrieved 25 March 2024.
54. SYDNEY J. FREEDBERG JR (13 June 2023). "Dumb and cheap: When facing electronic warfare in Ukraine, small drones' quantity is quality" (<https://breakingdefense.com/2023/06/dumb-and-cheap-when-facing-electronic-warfare-in-ukraine-small-drones-quantity-is-quality/>). *Breaking Defense*.
55. JOSEPH TREVITHICK; TYLER ROGOWAY (8 March 2024). "Russia Now Looks To Be Using Wire-Guided Kamikaze Drones In Ukraine" (<https://www.twz.com/air/russia-now-looks-to-be-using-wire-guided-kamikaze-drones-in-ukraine>). *TWZ*.
56. Mittal, Vikram. "Ukraine And Russia Battle To Defeat 'Un-jammable' Fiber-Optic Drones" (<https://www.forbes.com/sites/vikrammittal/2025/01/19/ukraine-and-russia-battle-to-defeat-un-jammable-fiber-optic-drones/>). *Forbes*. Retrieved 26 January 2025.
57. Panella, Chris. "Russia appears to be using wired, unjammable fiber-optic drones that could fix a big problem its operators have faced in this war" (<https://www.businessinsider.com/russia-fiber-optic-unjammable-drones-fix-a-big-problem-war-2024-8>). *Business Insider*. Retrieved 11 January 2025.
58. Hambling, David. "New Drone Tactics Sealed Russian Victory In Kursk" (<https://www.forbes.com/sites/davidhambling/2025/03/17/new-drone-tactics-sealed-russian-victory-in-kursk/>). *Forbes*. Retrieved 19 March 2025.
59. "Institute for the Study of War" (<https://www.understandingwar.org/backgrounder/russian-offensive-campaign-assessment-october-30-2024>). *Institute for the Study of War*. Retrieved 31 October 2024. "A spokesperson of a Ukrainian brigade operating in the Zaporizhia Oblast reported on October 29 that Russian forces began using first person view drones with fiber optic wires stretching up to 10 kilometers in length against Ukrainian forces in the Zaporizhia direction. The spokesperson added that these drones are reportedly immune to Ukrainian electronic warfare (EW) systems and that Ukrainian personnel had to shoot down these drones with small arms. The spokesperson observed that these drones are susceptible to air defense systems, have greatly limited ranges, and are ineffective in dense urban settings."
60. Barnes, Joe (10 January 2025). "How Russia created 'unjammable' drone with a retro twist" (<https://www.telegraph.co.uk/world-news/2025/01/10/russia-fibre-optic-drone-ukraine-war/>). *The Telegraph*. ISSN 0307-1235 (<https://search.worldcat.org/issn/0307-1235>). Retrieved 11 January 2025.
61. O'Grady, Siobhán; Khudov, Kostiantyn; Korolchuk, Serhii; Horton, Alex; Burianova, Tetiana; Morgunov, Serhiy; Karklis, Laris (23 May 2025). "Ukraine scrambles to overcome Russia's edge in fiber-optic drones" (<https://www.washingtonpost.com/world/2025/05/23/ukraine-russia-drones-fiberoptic-jamming/>). *The Washington Post*. ISSN 0190-8286 (<https://search.worldcat.org/issn/0190-8286>). Retrieved 27 May 2025.
62. THOMAS NEWDICK; TYLER ROGOWAY (21 March 2024). "Automated Terminal Attack Capability Appears To Be Making Its Way Into Ukraine's FPV Drones" (<https://www.twz.com/news-features/automated-terminal-attack-capability-appears-to-be-making-its-way-into-ukraines-fpv-drones>). *TWZ*.

63. "T-72M1 with a roof screen" (<https://twitter.com/naalsio26/status/1684559477549199360>). *Twitter*.
64. Boyko Nikolov (8 May 2023). "Russia is already putting ERAs on the tanks anti-drone grids" (<https://bulgarianmilitary.com/amp/2023/05/08/russia-is-already-putting-eras-on-the-tanks-anti-drone-grids/>). *bulgarianmilitary.com*. Retrieved 11 May 2023.
65. STETSON PAYNE (6 May 2023). "Automated Terminal Attack Capability Appears To Be Making Its Way Into Ukraine's FPV Drones" (<https://www.thedrive.com/the-war-zone/russian-tank-debuts-cope-cage-covered-in-explosive-reactive-armor>). *thedrive.com*. Retrieved 11 May 2023.
66. McFadden, Christopher (21 March 2024). "Ukraine UAV attacks force Russia to fit anti-drone cages on submarines" (<https://interestingengineering.com/military/russian-submarines-getting-anti-drone-cages>). *Interesting Engineering*.
67. Gilli, Andrea, and Mauro Gilli (2015). "The Diffusion of Drone Warfare? Industrial, Organizational and Infrastructural Constraints: Military Innovations and the Ecosystem Challenge" (<https://www.tandfonline.com/doi/full/10.1080/09636412.2016.1134189>), pages 21–22, 25–31
68. "Raytheon AGM/BGM/RGM/UGM-109 Tomahawk" (<https://www.designation-systems.net/dusrm/m-109.html>). *www.designation-systems.net*. Retrieved 18 July 2025.
69. RGM/UGM-109 Tomahawk (<http://www.fi-aeroweb.com/Defense/Tomahawk.html>), fi-aeroweb, November 2014
70. MQ-1B Predator factsheet (<https://web.archive.org/web/20111216191110/http://www.af.mil/information/factsheets/factsheet.asp?fsID=122>), US Air Force, July 2010
71. Kris Osborn, 18 May 2016, The U.S. Navy Has Big Plans for the Lethal Tomahawk Missile (<http://nationalinterest.org/blog/the-buzz/the-us-navy-has-big-plans-the-lethal-tomahawk-missile-16261?page=2>), The National Interest.
72. Takahashi, Timothy, et al. "A multi-disciplinary survey of advanced subsonic tactical cruise missile configurations." 43rd AIAA Aerospace Sciences Meeting and Exhibit. 2005.
73. Franklin, Michael (September 2008). "Unmanned combat air vehicles: opportunities for the guided weapons industry?" (https://web.archive.org/web/20180512181839/https://rusi.org/sites/default/files/200808_op_unmanned_combat_air_vehicles.pdf) (PDF). *RUSI Occasional Paper*. Archived from the original (https://rusi.org/sites/default/files/200808_op_unmanned_combat_air_vehicles.pdf) (PDF) on 12 May 2018. Retrieved 15 June 2017.
74. Liu, Xuancen (2016). "Optimal Design of Loitering Munition Trajectory in Complex Battlefield Environment" (http://www.atlantispress.com/php/download_paper.php?id=25855976). et al. Atlantis Press.
75. Snyder, Derek J. (25 April 2017). "Design Requirements for Weaponizing Man-Portable UAS in Support of Counter-Sniper Operations" (<https://web.archive.org/web/20170410050836/http://oai.dtic.mil/oai/oai?verb=getRecord&metadataPrefix=html&identifier=ADA551913>). Defense Technical Information Center. Archived from the original (<http://oai.dtic.mil/oai/oai?verb=getRecord&metadataPrefix=html&identifier=ADA551913>) on 10 April 2017.
76. Garcia, Denise. "Killer robots: Why the US should lead the ban." *Global Policy* 6.1 (2015): 57–63.
77. Sharkey, Noel. "Saying 'no!' to lethal autonomous targeting." *Journal of Military Ethics* 9.4 (2010): 369–383.
78. "Mortensen, Erika Steinholt. Autonomous weapon systems that decide whom to kill. How international humanitarian law and international human rights law regulate the development and use of offensive autonomous weapon systems during international armed conflicts. MS thesis. UiT Norges arktiske universitet, 2016" (<http://munin.uit.no/bitstream/handle/10037/9931/thesis.pdf?sequence=2&isAllowed=y>) (PDF).
79. Alston, Philip. "Lethal robotic technologies: the implications for human rights and international humanitarian law." *JL Inf. & Sci.* 21 (2011): 35.
80. Mayer, Chris; Williams, Andrew. "Crootof, Rebecca. "The Varied Law of Autonomous Weapon Systems." (2015)" (https://www.academia.edu/download/38978331/Autonomous-Systems-Publication_Print.pdf#page=120) (PDF). Retrieved 29 June 2024.
81. Evan Wallach and Erik Thomas (2016). "The Economic Calculus of Fielding Autonomous Fighting Vehicles Compliant with the Laws of Armed Conflict" (https://web.archive.org/web/20170410050957/http://yjolt.org/sites/default/files/Wallach_18YJoLT1.pdf) (PDF). *Yale Journal of Law & Technology*. 18: 1–25. Archived from the original (http://yjolt.org/sites/default/files/Wallach_18YJoLT1.pdf) (PDF) on 10 April 2017. Retrieved 9 April 2017.
82. Rivas, Santiago (27 December 2022). "Argentina compra municiones merodeadoras a Israel" (<https://www.pucara.org/post/argentina-compra-municiones-merodeadoras-a-israel>). *Pucará Defensa*.
83. "Armenian manufacturer boasts cutting-edge multifunctional combat UAVs and loitering munitions" (<https://armenpress.am/eng/news/928341.html>). *Armen press*. AM. 31 March 2018. Retrieved 19 July 2020.
84. EDEX 2023 witnesses AirWorker's AW21 loitering munition presentation (<https://www.airrecognition.com/index.php/news/defense-aviation-news/2023-news-aviation-aerospace/december/9424-edex-2023-witnesses-airworker-s-aw21-loitering-munition-presentation.html>) Archived (<https://web.archive.org/web/20231209235446/https://www.airrecognition.com/index.php/news/defense-aviation-news/2023-news-aviation-aerospace/december/9424-edex-2023-witnesses-airworker-s-aw21-loitering-munition-presentation.html>) 9 December 2023 at the Wayback Machine. *Air Recognition*. 5 December 2023.
85. A drone with a can-doom attitude (<https://www.c4isrnet.com/unmanned/2019/06/05/a-drone-with-a-can-doom-attitude/>). *C4ISRNET*. 5 June 2019.
86. "STM'nin yerli kamikaze İHA'sı Kargu Azerbaycan'da görüldü" (<https://www.defenceturk.net/stmnin-yerli-kamikaze-ihasi-kargu-azerbaycanda-goruldu>) (in Turkish). 29 October 2020.
87. "İlk kez Libya'da kullanılmıştı! Bu kez Azerbaycan'da görüntülendi" (<https://www.cnnturk.com/dunya/ilk-kez-libyada-kullanilmisti-bu-kez-azerbaycanda-goruntulendi>) (in Turkish). *CNN Türk*. 28 September 2020.
88. "ADEX 2022: Smart Point unveils 'attack' UAVs" (<https://www.janes.com/defence-news/news-detail/adex-2022-smart-point-unveils-attack-uavs>). *Janes Information Services*. 10 September 2022. Archived (<https://web.archive.org/web/20220911123709/https://www.janes.com/defence-news/news-detail/adex-2022-smart-point-unveils-attack-uavs>) from the original on 11 September 2022.
89. "Belarus-made armed drone shot down in Ukraine" (<https://defence-blog.com/belarus-made-armed-drone-shot-down-in-ukraine/>). *Defence blog*. 11 July 2022.

90. Brazilian firm Mac Jee unveils exploding drone, with demo in months (<https://www.defensenews.com/unmanned/uas/2023/11/17/brazilian-firm-mac-jee-unveils-exploding-drone-with-demo-in-months/>). *Defense News*. 17 November 2023.
91. DSA 2016: China details CH-901 UAV and loitering munition (<http://www.janes.com/article/59623/dsa-2016-china-details-ch-901-uav-and-loitering-munition>), Jane's, April 2016
92. Idex 2017: Catic reveals details about Harpy-type loitering munition (<http://www.janes.com/article/68341/idex-2017-catic-reveals-details-about-harpy-type-loitering-munition>), Jane's, March 2017
93. WB GROUP unmanned systems for Georgia (<http://www.wbgroup.pl/en/aktualnosci/wb-unmanned-systems-for-georgia/>). *WB Group*. 18 May 2022.
94. Defense Forces of Georgia Received UAVs Produced in Georgia and American Radars (<http://mod.gov.ge/en/news/read/9320/defense-forces-of-georgia-received-uavs-produced-in-georgia-and-american-radars>). *Ministry of Defence of Georgia*. 14 December 2023.
95. DEFEA 2023 – Attalus, a Greek Loitering Munition from Intracom Defense (<https://www.edrmagazine.eu/attalus-a-greek-loitering-munition-from-intracom-defense>). *European Defence Review*. 12 May 2023.
96. EDEX 2023: Greek company SAS Technology displays Aihmi AHM-1X 70mm loitering munition (<https://www.airrecognition.com/index.php/news/defense-aviation-news/2023-news-aviation-aerospace/december/9431-edex-2023-greek-company-sas-technology-displays-aihmi-ahm-1x-70mm-loitering-munition.html>) Archived (<https://web.archive.org/web/20231209235445/https://www.airrecognition.com/index.php/news/defense-aviation-news/2023-news-aviation-aerospace/december/9431-edex-2023-greek-company-sas-technology-displays-aihmi-ahm-1x-70mm-loitering-munition.html>) 9 December 2023 at the Wayback Machine. *Air Recognition*. 6 December 2023.
97. "Gelar soft launching munition Bernama Rajata" (<https://web.archive.org/web/20230320194336/https://dahana.id/berita/dahana-gelar-soft-launching-loitering-munition-bernama-rajata/>). Dahana. Archived from the original (<https://dahana.id/berita/dahana-gelar-soft-launching-loitering-munition-bernama-rajata/>) on 20 March 2023. Retrieved 20 November 2022.
98. Yanwardhana, Emir. "Indonesia Produksi Drone 'Bunuh Diri', Begini Kekuatannya" (<https://www.cnbcindonesia.com/news/20221102192140-4-384686/indonesia-produksi-drone-bunuh-diri-begini-kekuatannya>). *CNBC Indonesia* (in Indonesian). Retrieved 5 February 2025.
99. "Army unleashes 'silent killer' indigenous Nagastra-1 suicide drones along Pakistan, China borders: Key features revealed" (<https://economictimes.indiatimes.com/news/defence/army-unleashes-silent-killer-indigenous-nagastra-1-suicide-drones-along-pakistan-china-borders-key-features-revealed/articleshow/110996803.cms?from=mdr>). *The Economic Times*. 14 June 2024. ISSN 0013-0389 (<https://search.worldcat.org/issn/0013-0389>). Retrieved 15 June 2024.
100. Harop Loitering Munitions System for the IAF (<http://www.indiandefencereview.com/news/harop-loitering-munitions-system-for-the-iaf/>), India Defence Review, January 2014
101. "Laser weapons, swarm drones on DRDO menu" (<https://www.thehindu.com/sci-tech/science/drdo-focus-on-stealth-weapons-drones/article28988571.ece>). *The Hindu*. 12 August 2019. ISSN 0971-751X (<https://search.worldcat.org/issn/0971-751X>). Retrieved 11 August 2020.
102. Pubby, Manu (22 September 2022). "Indigenous loitering munition successfully hits target at Pokhran" (<https://economictimes.indiatimes.com/news/defence/indigenous-loitering-munition-successfully-hits-target-at-pokhran/articleshow/94383125.cms>). *The Economic Times*. New Delhi. Retrieved 23 September 2022.
103. "India's first kamikaze drone developed; 5,000 units can be made in 2-3 yrs" (<https://www.theweek.in/news/india/2024/05/08/indias-first-kamikaze-drone-developed-5000-units-can-be-made-in-2-3-yrs.html>). *The Week*. Retrieved 9 May 2024.
104. Sof, Eric (20 October 2022). "HESA Shahed 136: A cheap and deadly Iranian kamikaze drone" (<https://special-ops.org/hesa-shahed-136-kamikaze-drone/>). *Spec Ops Magazine*. Retrieved 1 December 2023.
105. "Russia's Iranian-Made UAVs: A Technical Profile" (<https://www.rusi.org/explore-our-research/publications/commentary/russia-iranian-made-uavs-technical-profile>). *Royal United Services Institute*. 13 January 2023.
106. Iran's navy touts 'suicide drone' (<https://www.usatoday.com/story/news/world/2016/10/26/irans-navy-unveils-suicide-drone/92761598/>), USA Today, October 2016
107. Iran Tests 'Kamikaze' Suicide Drone (<http://nationalinterest.org/blog/iran-tests-kamikaze%E2%80%99-suicide-drone-11934>), The National Interest Papers Offer a Peek at ISIS' Drones, Lethal and Largely Off-the-Shelf (<https://www.nytimes.com/2017/01/31/world/middleeast/isis-drone-documents.html>), New York Times, January 2017
108. Portable Attack Drones or Loitering Munitions (<https://www.pressreader.com/india/sps-landforces/20160915/281779923685619>), SPS Landforces, September 2016
109. US Switchblade Drone 'Knock Off' Meraj-521 Unveiled By Iran; Comes After Grand Success Of Its UAVs In Ukraine (<http://eurasianimes.com/us-switchblade-drone-knock-off-meraj-521-unveiled-by-iran-comes/>). *EurAsian Times*. 21 October 2022.
110. Aeronautics introduces Orbiter 1K loitering munition (<https://www.flightglobal.com/news/articles/aeronautics-introduces-orbiter-1k-loitering-munition-412651/>), Flight Global, May 2015
111. Elbit announces new SkyStriker loitering munition (<http://www.janes.com/article/63526/elbit-announces-new-skystriker-loitering-munition>), Jane's Defence Weekly, September 2016
112. Israeli companies pitching loitering munitions for US Army programme (<https://www.flightglobal.com/news/articles/israeli-companies-pitching-loitering-munitions-for-u-423808/>), FlightGlobal, April 2016
113. Soyulu, Ragip (25 March 2021). "US Army, Marines, Special Forces Eye Israeli 'Hero' Attack Drones" (<https://web.archive.org/web/20230219153159/https://breakingdefense.sites.breakingmedia.com/2021/03/us-army-marines-special-forces-eye-israeli-hero-attack-drones/>). *Breaking Defense*. Archived from the original (<https://breakingdefense.sites.breakingmedia.com/2021/03/us-army-marines-special-forces-eye-israeli-hero-attack-drones/>) on 19 February 2023. Retrieved 1 September 2021.
114. Trevithick, Joseph (23 June 2021). "Marines Pick Loitering Munition To Arm Light Vehicles And Drone Boats" (<https://www.the-war-zone.com/41219/marines-pick-loitering-munition-to-arm-light-vehicles-and-drone-boats>). *The Drive*. Retrieved 1 September 2021.

115. "Israeli UVision to Supply Hero-120 Loitering Munition to U.S. Marine Corps" (https://web.archive.org/web/20211026154634/https://www.defenseworld.net/news/29856/Israeli_UVision_to_Supply_Hero_120_Loitering_Munition_to_U_S_Marine_Corps). *www.defenseworld.net*. 21 June 2021. Archived from the original (https://www.defenseworld.net/news/29856/Israeli_UVision_to_Supply_Hero_120_Loitering_Munition_to_U_S_Marine_Corps) on 26 October 2021. Retrieved 1 September 2021.
116. "SpearUAV unveils Viper micro-tactical loitering munition" (<https://www.janes.com/defence-news/news-detail/spearuav-unveils-viper-micro-tactical-loitering-munition>). *Janes Information Services*. 6 October 2022. Archived (<https://web.archive.org/web/20221018183931/https://www.janes.com/defence-news/news-detail/spearuav-unveils-viper-micro-tactical-loitering-munition>) from the original on 18 October 2022.
117. AI-driven combat drone can search buildings and execute suicide attacks (<https://newatlas.com/military/lanius-ai-suicide-drone-bomb/>). *New Atlas*. 20 November 2022.
118. Israel Aerospace Industries unveils Point Blank – a hand-launched electro-optical guided missile (<https://www.edrmagazine.eu/israel-aerospace-industries-unveils-point-blank-a-hand-launched-electro-optical-guided-missile>). *European Defence Review*. 19 January 2023.
119. PARIS AIR SHOW NEWS: New Loitering Munition Promises to Go the Distance (<https://www.nationaldefensemagazine.org/articles/2023/6/20/new-loitering-munition-promises-to-go-the-distance>). *National Defense Magazine*. 20 June 2023.
120. Vietnam Eyes Israel's Delilah Standoff Missile, and F-16s Could Be Next (<http://www.thedrive.com/the-war-zone/8219/vietnam-eyes-israels-delilah-standoff-missile-and-f-16s-could-be-next>). *The Warzone*, March 2017
121. "'First in the world': Lithuania signs contract to acquire Switchblade combat drones from US" (<https://www.lrt.lt/en/news-in-english/191851474/first-in-the-world-lithuania-signs-contract-to-acquire-switchblade-combat-drones-from-us>). *LRT*. 22 December 2022. Retrieved 22 December 2022.
122. Warmate expendable UAV in production for two customers (<https://www.flightglobal.com/news/articles/warmate-expendable-uav-in-production-for-two-custome-424735/>), *Flight Global*, April 2016
123. *Vai, ataca alvos a 200 km/h, e já não volta. Elanus, o primeiro drone armado português* (<https://cnnportugal.iol.pt/videos/vai-i-ataca-alvos-a-200-km-h-e-ja-nao-volta-elanus-o-primeiro-drone-armado-portugues/650d8b660cf25f9953837686>) (News) (in Portuguese). *CNN Portugal*. Retrieved 24 September 2023.
124. Под крышей дрона: ВМФ усиливается БПЛА-камикадзе (<https://iz.ru/1242367/aleksei-ramm-bogdan-stepovoi/pod-krysh-ei-drona-vmf-usilivaetsia-bpla-kamikadze?destination=node/1242367>), *Izvestia*, 20 October 2021
125. Russian Naval Ships to Be Armed With Kamikaze Drones (<https://www.thedefensepost.com/2021/11/02/russia-ships-kamikaze-drones/>), *The Defense Post*, 2 November 2021
126. "Russia Attacked U.S. Supplied Howitzers in Ukraine With Kamikaze Drones and Rockets" (<https://www.19fortyfive.com/2022/05/russia-attacked-u-s-supplied-howitzers-in-ukraine-with-kamikaze-drones-and-rockets/>). 19 May 2022.
127. Serbia may become biggest operator of military drones in Balkans (<https://www.defensenews.com/unmanned/2022/11/21/s-erbia-may-become-biggest-operator-of-military-drones-in-balkans/>). *Defense News*. 21 November 2022.
128. Serbia's OSICA loitering munition underlines rising relevance of such system (https://www.armyrecognition.com/defense_news_september_2023_global_security_army_industry/serbia_s_osica_loitering_munition_underlines_rising_relevance_of_such_system.html) Archived (https://web.archive.org/web/20240106195933/https://www.armyrecognition.com/defense_news_september_2023_global_security_army_industry/serbia_s_osica_loitering_munition_underlines_rising_relevance_of_such_system.html) 6 January 2024 at the *Wayback Machine*. *Army Recognition*. 29 September 2023.
129. Yugoimport develops Raven 145 loitering area denial weapon system (https://www.armyrecognition.com/defense_news_october_2023_global_security_army_industry/yugoimport_develops_raven_145_loitering_area_denial_weapon_system.htm). *Army Recognition*. 3 October 2023.
130. Discover Vila 1 modular loitering munition from Serbian Vlatacom Institute (https://www.armyrecognition.com/defense_news_october_2023_global_security_army_industry/discover_the_vila_1_modular_loitering_munition_from_serbian_vlatacom_institute.html). *Army Recognition*. 3 October 2023.
131. Martin, Guy (22 February 2023). "Paramount putting N-Raven loitering munition into production" (<https://www.defenceweek.co.za/aerospace/aerospace/paramount-putting-n-raven-loitering-munition-into-production/>). *defenceWeb*. Retrieved 15 March 2023.
132. South Korea's Kamikaze UAV Could Scare the Ojom Out of Kim Jong-un (<https://gizmodo.com/5950970/south-koreas-kamikaze-uav-aims-to-ruin-kim-jong-uns-week>) Archived (<https://web.archive.org/web/20171012202319/https://gizmodo.com/5950970/south-koreas-kamikaze-uav-aims-to-ruin-kim-jong-uns-week>) 12 October 2017 at the *Wayback Machine*, *Gizmodo*, October 2012
133. South Korea developing 'kamikaze' attack drone (<https://www.foxnews.com/tech/south-korea-developing-kamikaze-attack-drone/>), *Fox News*, October 2012
134. Wong, Kelvin. "TADTE 2019: Taiwan's NCSIST rolls out indigenous anti-radiation loitering munition" (<https://www.janes.com/article/90511/tadte-2019-taiwan-s-ncsist-rolls-out-indigenous-anti-radiation-loitering-munition>). *Jane's*. Retrieved 21 October 2019.
135. "Robit" (<http://robit.com.tr>).
136. IAI Gets \$100 Million Contract for HAROP Killer Drones (http://defense-update.com/newscast/0609/news/harop_new_140609.html) Archived (https://web.archive.org/web/20170720115709/http://www.defense-update.com/newscast/0609/news/harop_new_140609.html) 20 July 2017 at the *Wayback Machine*, *Defense Update*, 2009
137. "Turkey's STM delivering Kargu loitering munitions to TSK" (<https://www.janes.com/article/88686/turkey-s-stm-delivering-kargu-loitering-munitions-to-tsk>). *Jane's 360*. Retrieved 23 March 2020.
138. "ALPAGU® Fixed Wing Loitering Munition System" (<https://www.stm.com.tr/en/alpagu>). *STM*. TR.
139. "Turkish firms develop kamikaze drone designed to neutralise drone-borne threats" (<https://www.unmannedairspace.info/content/under-uas-systems-and-policies/turkish-firms-develop-kamikaze-drone-designed-to-neutralise-drone-borne-threats/>). *Unmanned air space*. 29 October 2021.

140. "Lentatek Unveils Kargi Anti-Radiation UAV For the First Time" (<https://web.archive.org/web/20221021154834/https://www.turkishdefencenews.com/lentatek-unveils-kargi-anti-radiation-uav-for-the-first-time/>). *Turkish Defence news*. 6 June 2022. Archived from the original (<https://www.turkishdefencenews.com/lentatek-unveils-kargi-anti-radiation-uav-for-the-first-time/>) on 21 October 2022. Retrieved 11 July 2022.
141. "STM and Roketsan from Türkiye unveil new Alpagut loitering munition" (https://www.armyrecognition.com/defense_news_october_2022_global_security_army_industry/stm_and_roketsan_from_turkiye_unveil_new_alpagut_loitering_munition.html). *Army recognition*. 28 October 2022.
142. Turkish firms unveil a new loitering munition (<https://www.defensenews.com/global/europe/2022/10/28/turkish-firms-unveil-a-new-loitering-munition/>). *Defense News*. 28 October 2022.
143. ADASI EDGE presents QX-1 new loitering munition fully developed in UAE (https://www.armyrecognition.com/weapons_defence_industry_military_technology_uk/adasi_edge_presents_qx-1_new_loitering_munition_fully_developed_in_uae.html). *Army Recognition*. 9 April 2021.
144. Edge Group unveils kamikaze drones at Idex (<https://www.defensenews.com/digital-show-dailies/idex/2021/02/22/edge-group-unveils-kamikaze-drones-at-idex/>). *Defense News*. 22 February 2021.
145. Hunter SP (<https://edgegroup.ae/solutions/hunter-2-single-tube-launched-drone-ground>). *EDGE Group*.
146. UAE Shows Off A Compact Box Launcher Concept Full Of 21 AI-Enabled Swarming Suicide Drones (<https://www.thedrive.com/the-war-zone/44395/uae-shows-off-a-compact-box-launcher-concept-full-of-21-ai-enabled-swarming-suicide-drones>). *The Drive/The War Zone*. 21 February 2022.
147. Following first demonstration, Edge unveils swarming drones based on AI technology (<https://www.defensenews.com/industry/2022/02/22/following-first-demonstration-edge-unveils-swarming-drones-based-on-ai-technology/>). *Defense News*. 22 February 2022.
148. "Dubai Airshow 2021: Halcon unveils Hunter family of ISR UAVs" (<https://www.janes.com/defence-news/news-detail/dubai-airshow-2021-halcon-unveils-hunter-family-of-isr-uavs>). *Janes Information Services*. 17 November 2021. Archived (<https://web.archive.org/web/20211117154825/https://www.janes.com/defence-news/news-detail/dubai-airshow-2021-halcon-unveils-hunter-family-of-isr-uavs>) from the original on 17 November 2021.
149. Halcon Edge from UAE has designed Shadow jet-engine powered UAVs loitering munitions (https://www.armyrecognition.com/weapons_defence_industry_military_technology_uk/halcon_edge_from_uae_has_designed_shadow_jet-engine_powered_uavs_loitering_munitions.html). *Army Recognition*. 9 April 2021.
150. As UAV tech spreads, Gulf firms bet on upgrades with drone-to-satellite links (<https://breakingdefense.com/2022/06/as-uav-tech-spreads-gulf-firms-bet-on-upgrades-with-drone-to-satellite-links/>). *Breaking Defense*. 3 June 2022.
151. HALCON from UAE unveils RW-24 Smart Loitering Precision Attack munitions (https://www.armyrecognition.com/defense_news_may_2021_global_security_army_industry/adasi_from_uae_unveils_rw-24_smart_loitering_precision_attack_munitions.html). *Army Recognition*. 10 May 2021.
152. Paramount Aerospace Systems to start this year production of N-Raven loitering munition (https://www.armyrecognition.com/defense_news_february_2023_global_security_army_industry/paramount_aerospace_systems_to_start_this_year_production_of_n-raven_loitering_munition.html). *Army Recognition*. 27 February 2023.
153. Surface Navy 2017: Coyote earmarked for ISR and offensive roles (<http://www.janes.com/article/66954/surface-navy-2017-coyote-earmarked-for-isr-and-offensive-roles>), Jane's, January 2017
154. "Lumberjack Jet-Powered One-Way Attack Munition Can Drop Its Own Precision Bomblets (Updated)" (<https://www.twz.com/air/lumberjack-jet-powered-one-way-attack-munition-can-drop-its-own-precision-bomblets>). 29 April 2025.
155. "Ukrainian AN-196 Liutyi Drone, Dubbed 'Ukrainian Shahed,' Receives Major Upgrade" (<https://united24media.com/latest-news/ukrainian-an-196-liutyi-drone-dubbed-ukrainian-shahed-receives-major-upgrade-3996>). 25 November 2024.
156. Suicide Drones Have Migrated to The Conflict in Yemen (<http://www.thedrive.com/the-war-zone/8586/suicide-drones-have-migrated-to-the-conflict-in-yemen>), The Warzone, March 2017
157. "North Korea's Kim Jong Un oversees tests of new AI-equipped suicide drones" (<https://www.aljazeera.com/news/2025/3/27/north-koreas-kim-jong-un-oversees-tests-of-new-ai-equipped-suicide-drones>). *Al Jazeera*. Retrieved 1 April 2025.

Further reading

- Voskuil, Mark (March 2022). "Performance analysis and design of loitering munitions: A comprehensive technical survey of recent developments" (<https://doi.org/10.1016%2Fj.dt.2021.08.010>). *Defence Technology*. **18** (3): 325–343. doi:10.1016/j.dt.2021.08.010 (<https://doi.org/10.1016%2Fj.dt.2021.08.010>). ISSN 2214-9147 (<https://search.worldcat.org/isbn/2214-9147>).

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